

Arunabha Majumder

Mechanical Engineer — Robotic Actuators, Joint Kinematics, Prototyping & Test Rigs

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PROJECT NOTE: HYBRID-ACTUATED ROBOTIC SHOULDER JOINT

What I built

A robotic shoulder joint for an exoskeleton (PhD project VIEXO, Aalborg University) that supports the arm against gravity. I designed the actuation concept, the mechanical structure and assemblies, made the parts, integrated the electronics and built the test rigs. Project page: www.viexo.aau.dk

Challenge 1: Torque demand vs. motor size and weight

Carrying the full gravity torque with a motor alone would need a large, heavy and power-hungry actuator. I combined a parallel spring that carries the quasi-static gravity load with a small 6 Nm motor that only supplies what the spring cannot. This kept the joint light and cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026).

Challenge 2: Strength and stiffness without adding weight

Load-carrying parts had to stay stiff and strong while remaining light. I used FEA to iterate the geometry for stiffness, strength and weight before manufacture, and chose between machined and 3D-printed parts depending on the part, so I could get a first prototype built quickly and improve from there.

Challenge 3: Getting reliable force information

On a related 5-bar planar parallel robot, I estimated the end-effector force from the variable-stiffness actuators through the robot's Jacobian and validated it against a force sensor to below 10% RMS error. I then used it to measure human arm impedance in the gait lab.

Challenge 4: Making the prototype work on the bench

I integrated the motor, encoder, force sensors, DAQ and embedded control, built custom test rigs, and tracked down mechanical, electrical and control issues through repeated test campaigns until the system ran reliably in tests with participants.

More projects and photos: arunabha22.github.io/arunabha-majumder